

if Conditional Branch

ELEC1006: ENGINEERING COMPUTING

Conditional Branches

- Defined as a condition that causes the program to change the path of execution based on the value of an expression.
- Forms of conditional branches:
 - *if* statement. ←
 - *switch* statement.

if Statement

- The '*if*' (a keyword) is to execute a statement or a block of statements only if a condition is satisfied. The form is:
 - *if* (condition) statement
 - e.g. *if* (x>90)
 cout << "A very good score!" << endl;

This 'cout statement' will only be executed if the value in x is greater than 90.

Multiple Lines of Code

- if multiple lines of code is to be executed under an *if* statement, {} should be used.
- Do **NOT** put ; after the *if (condition)* statement.
- e.g. **if** (x>90)

{

cout << "Congratulation!"<< endl;

cout << "A very good score of ";

cout << x << "." << endl;

}

Two Conditional Branches

- The **else** statement can be used to specify what to do if the 'if' condition is not met.
- e.g. **if** (x>90) **// Condition 1**
 cout << "Congratulation!"<<endl;
else **// Condition 2**
 cout << "You can improve your score";

Multiple Conditional Branches

- Use **if – else if – else** for a range of conditional branches.

– e.g.

```
if (x<0)           // Condition 1
```

```
    cout << "x is negative." << endl;
```

```
else if (x>0)      // Condition 2
```

```
    cout << "x is positive." << endl;
```

```
else              // Condition 3
```

```
    cout << "x is 0." << endl;
```

Commas in *if* Statement

- Use of commas ,

- e.g.

```
int a=1,b=2,c=10,d=20;
```

```
if (a<0) c*=10, d*=20;
```

```
else if (a<b) c/=10, d/=20;
```

```
else c=b;
```

Previous code is equivalent to:

```
int a=1;
int b=2;
int c=10;
int d=20;
if (a<0)
{
    c=c*10;
    d=d*20;
}
else if (a<b)
{
    c=c/10;
    d=d/20;
}
else
    c=b;
```

— Too long?

More info

- [1] cplusplus.com: Statements and flow control
<https://cplusplus.com/doc/tutorial/control/>
- [2] learncpp.com: 5.2 – If statements
<https://www.learncpp.com/cpp-tutorial/52-if-statements/>